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AIM:Feed Forward Backpropagation Neural Network

- Implement the Feed Forward Backpropagation algorithm to train a neural
- network.
- Use a given dataset to train the neural network for a specific task.
- Evaluate the performance of the trained network on test data.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")
df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)


df["Output"] = df["Output"].map({"Yes": 1, "No": 0})
X = pd.get_dummies(
    df.drop(["Output", "Feedback"], axis=1, errors="ignore"), drop_first=True
)
y = df["Output"]

X_scaled = StandardScaler().fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)


model = MLPClassifier(
    hidden_layer_sizes=(16, 8), max_iter=300, random_state=42
)
model.fit(X_train, y_train)


y_pred = model.predict(X_test)
print(f"Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%")
print(classification_report(y_test, y_pred))

plt.plot(model.loss_curve_)
plt.title("Backpropagation Loss Curve")
plt.show()
```

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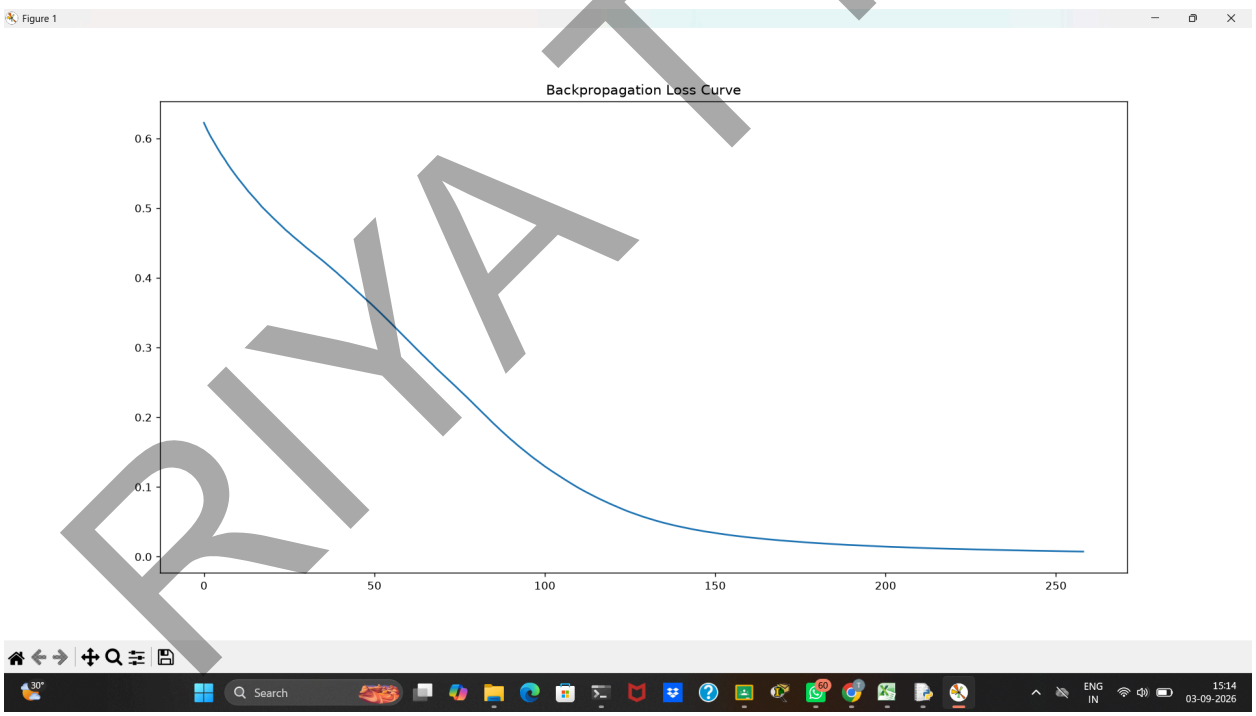
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OUTPUT:

```
*IDLE Shell 3.14.7*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\hrane\Documents\AI_R.py ==========
Accuracy: 97.44%
              precision    recall  f1-score   support

           0       0.89      1.00      0.94        17
           1       1.00      0.97      0.98        61

    accuracy                           0.97        78
   macro avg       0.95      0.98      0.96        78
weighted avg       0.98      0.97      0.97        78
```



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